



INSIGHT MINDS



AI-Powered Job Market Insights

Done by:

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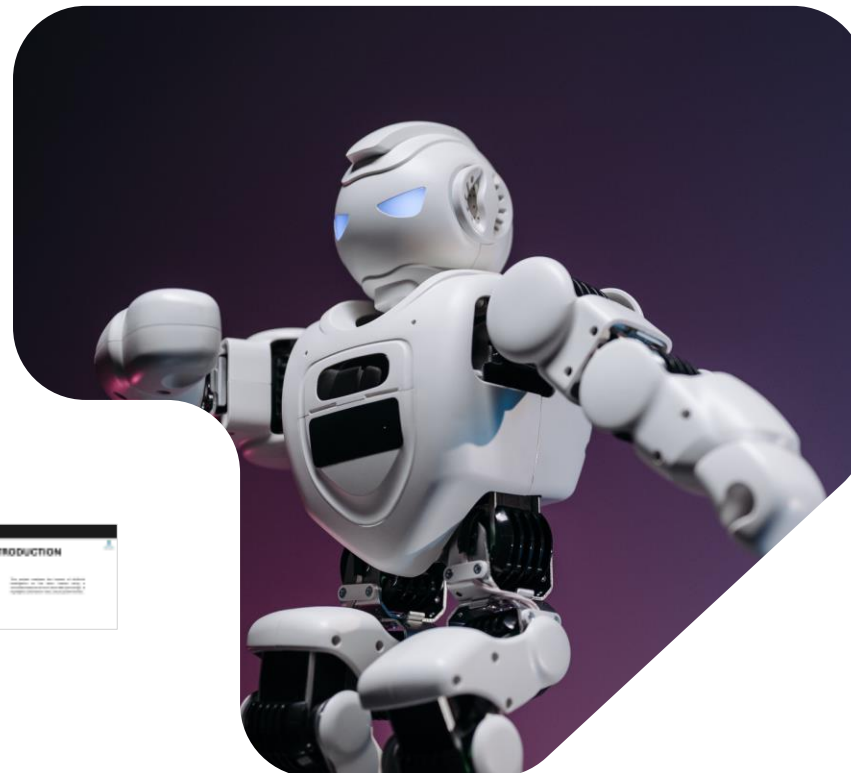
Sogoud Ahmed

Moataz Samir

Elsayed Abdellatef

Youssef Mohammed

Supervised by: DR. Amal Mahmoud





INTRODUCTION



This project analyzes the impact of Artificial Intelligence on the labor market using a simulated dataset of more than 500 job listings. It highlights automation risks, future growth trends.



OBJECTIVES

TRENDS

Explore employment trends influenced by AI.

JOBS

Identify vulnerable job categories at risk of automation.

INSIGHTS

Provide insights for policymakers, companies, and professionals.





METHODOLOGY



Dataset

- About Dataset
- Dataset Description



Data Preprocessing

- Data Preparation.
- Data Cleaning & Preprocessing.
- Data modeling



Data Analysis

- EDA
- Data visualization



DATASET

About Dataset

This dataset was extracted from Kaggle.com,
[AI Impact on Job Market: \(2024–2030\)](#)

DATASET DESCRIPTION

Total Records: 30k
Total Columns: 13

Job Title	Industry	Job Status	AI Impact	Median Salary	Expected Unemployment	Expected	Job Openings	Projected Supply	Industry Workforce	Automation Risk (%)	Location	Gender Diversity (%)
Accountant Analyst	IT	Increasing	Medium	42186.75	Medium-High Degree	9	1810	8342	95.96	36.29	US	44.63
Architect, landscape	Architectural/Engineering	Medium	Low	232296.17	Post-Grad Degree	10	1242	6205	19.83	89.71	USA	96.59
Business Development	Finance	Increasing	Low	143276.18	Postgraduate Degree	8	3936	2158	62.82	75.87	Canada	81.13
Case manager	Healthcare	Increasing	High	67576.12	Associate Degree	20	7172	4950	1.89	90.94	Australia	93.70
Chemical Engineer	IT	Increasing	Low	80698.60	Postgraduate Degree	20	8648	7280	33.76	87.69	Germany	72.87
Legal secretary	Education	Decreasing	Low	95121.51	Master's Degree	20	6905	7612	77.4	30.09	USA	48.93
Scrum master	Healthcare	Decreasing	High	147186.00	High School	7	8730	4300	80.91	1.06	US	26.34
Teacher	Education/Healthcare	Decreasing	High	64248.40	Associate Degree	8	1392	8305	9.74	2.40	Germany	93.77
Emergency planning/management	Education	Increasing	High	72488.23	High School	20	7700	1458	26.71	89.16	Australia	96.86
Visual merchandiser	Healthcare	Decreasing	Medium	68440.3	Associate Degree	18	1107	2175	4.79	89.03	China	44.32
Emergency planning/management	Healthcare	Increasing	Medium	81123.62	Postgraduate Degree	17	8627	2695	99.99	49.94	China	34.63
Accounting manager/analyst	Finance	Decreasing	High	114870.3	High School	10	2900	7664	96.15	30.13	US	81.9
Biological scientist	Healthcare	Increasing	Medium	330427.00	PhD	7	1802	3700	52.89	0.54	Australia	36.02
Business Development Manager	Healthcare	Decreasing	Medium	43089.64	PhD	2	1814	7626	88.6	10.12	Australia	77.6
Product development	Healthcare	Decreasing	High	237994.24	Postgraduate Degree	8	8914	8760	59.46	91.79	India	39.96
Business analyst	Healthcare	Increasing	Medium	87868.86	Postgraduate Degree	3	1870	8314	93.9	0.19	China	74.67
Team player	Health	Decreasing	High	187208.60	Associate Degree	12	7911	2960	15.25	95.15	US	29.99
Engineer (electrical/electronic)	Healthcare	Decreasing	Medium	181233.00	Associate Degree	17	1246	8829	12.82	84.18	India	98.12
Biological scientist	Healthcare	Increasing	Low	200546.90	Postgraduate Degree	8	5170	8613	60.12	36.02	US	36.07
General scientist, biological	Healthcare	Decreasing	High	142088.31	Postgraduate Degree	3	2877	7807	6.29	98.68	India	96.17
Product development	Healthcare	Increasing	High	107599.20	High School	4	2981	5698	89.17	26.3	India	34.04
Advertising manager	Health	Decreasing	High	122688.88	Postgraduate Degree	16	7281	7409	60.84	1.16	India	74.43
Economist/business analyst	Healthcare	Decreasing	Low	69374.45	Associate Degree	14	5802	395	40.78	89.98	US	96.9
Secretary/business assistant	Healthcare	Increasing	Medium	48770.11	PhD	4	5200	8021	20.77	86.11	Germany	96.86
Product development, public	Healthcare	Increasing	Low	89003.01	High School	1	9400	3832	86.18	40.07	China	79.99



DATASET DESCRIPTION

Total Records: 30k

Total Columns: 13

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Job Title	Industry	Job Status	AI Impact	Median Salary	Required Education	Experience	Job Openings	Projected Openings	Remote Work Rate	Automation Risk (%)	Location	Gender Diversity (%)
2	Investment analyst	IT	Increasing	Moderate	42109.76	Master's Degree	5	1515	6342	55.96	28.28	UK	44.63
3	Journalist, newspaper	Manufacturing	Increasing	Moderate	132298.57	Master's Degree	15	1243	6205	16.81	89.71	USA	66.39
4	Financial planner	Finance	Increasing	Low	143279.19	Bachelor's Degree	4	3338	1154	91.82	72.97	Canada	41.13
5	Legal secretary	Healthcare	Increasing	High	97576.13	Associate Degree	15	7173	4060	1.89	99.94	Australia	65.76
6	Aeronautical engineer	IT	Increasing	Low	60956.63	Master's Degree	13	5944	7396	53.76	37.65	Germany	72.57
7	Legal secretary	Education	Decreasing	Low	39123.32	Master's Degree	20	6985	7832	77.5	28.99	USA	46.91
8	Surveyor, insurance	Manufacturing	Decreasing	High	147150.03	High School	7	9738	4360	30.51	7.08	UK	20.14
9	Dentist	Entertainment	Decreasing	High	64245.48	Associate Degree	0	1393	8206	9.74	2.49	Canada	59.27
10	Emergency planning/management	Education	Increasing	High	72488.15	High School	12	7729	4452	38.77	99.56	Australia	60.56
11	Visual merchandiser	Manufacturing	Decreasing	Moderate	95840.3	Associate Degree	16	1127	2375	4.29	69.03	China	48.19
12	Emergency planning/management	Manufacturing	Increasing	Moderate	97133.62	Bachelor's Degree	17	4627	1684	89.49	43.94	China	34.83
13	Housing manager/officer	Entertainment	Decreasing	High	114672.3	High School	12	2800	7964	94.02	20.19	UK	31.9
14	Geologist, wellsite	Entertainment	Increasing	Moderate	116427.68	PhD	7	1852	3700	32.89	5.54	Australia	56.02
15	Runner, broadcasting/film/video	Healthcare	Decreasing	Moderate	43059.04	PhD	2	1824	7625	55.6	13.12	Australia	27.5
16	Press photographer	Entertainment	Decreasing	Moderate	137894.14	Master's Degree	3	8312	9700	59.95	95.76	India	56.06
17	Biochemist, clinical	Entertainment	Increasing	Moderate	67908.84	Master's Degree	2	1872	5314	93.9	3.19	China	74.67
18	Town planner	Retail	Decreasing	High	107318.63	Associate Degree	11	7251	1090	15.25	55.15	UK	23.99
19	Engineer, civil (consulting)	Healthcare	Decreasing	Moderate	101322.09	Associate Degree	17	1044	8832	12.92	64.18	India	56.12
20	Audiological scientist	Transportation	Increasing	Low	109546.76	Master's Degree	8	5379	8913	82.13	80.62	UK	38.87
21	Clinical scientist, histocompatibility	Manufacturing	Decreasing	High	142660.31	Master's Degree	2	2377	7307	4.29	16.88	India	56.57
22	Music therapist	Education	Increasing	High	122569.29	High School	4	2661	5588	88.17	26.8	India	34.19
23	Advertising copywriter	Retail	Decreasing	High	122668.89	Bachelor's Degree	14	7291	7469	65.94	7.14	Brazil	74.45
24	Commercial horticulturist	Education	Decreasing	Low	60874.45	Associate Degree	14	5892	369	43.78	95.96	UK	56.8
25	Secondary school teacher	Finance	Increasing	Moderate	49705.11	PhD	4	5398	3531	20.77	60.11	Germany	49.88
26	Production assistant, radio	Healthcare	Increasing	Low	59923.23	High School	1	8039	3829	55.19	48.57	China	78.99

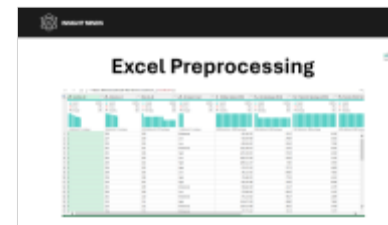
ai_job_trends_dataset



DATASET PREPROCESSING

1

We Handled Missing Data



2

Detected Duplicated Values



3

Data Type Correction





Excel Preprocessing

fx | = Table.RemoveColumns("#Reordered Columns2",{"Location"})

location_id	industry_id	job_id	AI Impact Level	Median Salary (USD)	Job Openings (2024)	Projected Openings (2030)	Remote Work Rat
Valid 100% Error 0% Empty 0% 8 distinct, 0 unique	Valid 100% Error 0% Empty 0% 8 distinct, 0 unique	Valid 100% Error 0% Empty 0% 828 distinct, 670 unique	Valid 100% Error 0% Empty 0% 3 distinct, 0 unique	Valid 100% Error 0% Empty 0% 1000 distinct, 1000 unique	Valid 100% Error 0% Empty 0% 948 distinct, 898 unique	Valid 100% Error 0% Empty 0% 953 distinct, 906 unique	Valid Error 0% Empty 0% 945 distinct, 890 unique
1	200	200	Moderate	42,109.76	1515	6342	
2	214	200	Low	41,997.68	2409	5015	
3	200	204	Low	60,956.63	5944	7396	
4	202	201	Moderate	132,298.57	1243	6205	
5	202	206	High	147,150.03	9738	4360	
6	204	202	Low	143,279.19	3338	1154	
7	200	202	High	109,421.37	985	3954	
8	206	203	High	97,576.13	7173	4060	
9	208	205	Low	39,123.32	6985	7832	
10	208	208	High	72,488.15	7729	4452	
11	210	207	High	64,245.48	1393	8206	
12	202	209	Moderate	95,840.30	1127	2375	
13	206	209	Low	37,084.34	9919	1155	
14	202	210	Moderate	97,133.62	4627	1684	
15	210	211	High	114,672.30	2800	7964	
16	210	212	Moderate	116,427.68	1852	3700	
17	206	213	Moderate	43,059.04	1824	7625	



Excel Dashboard



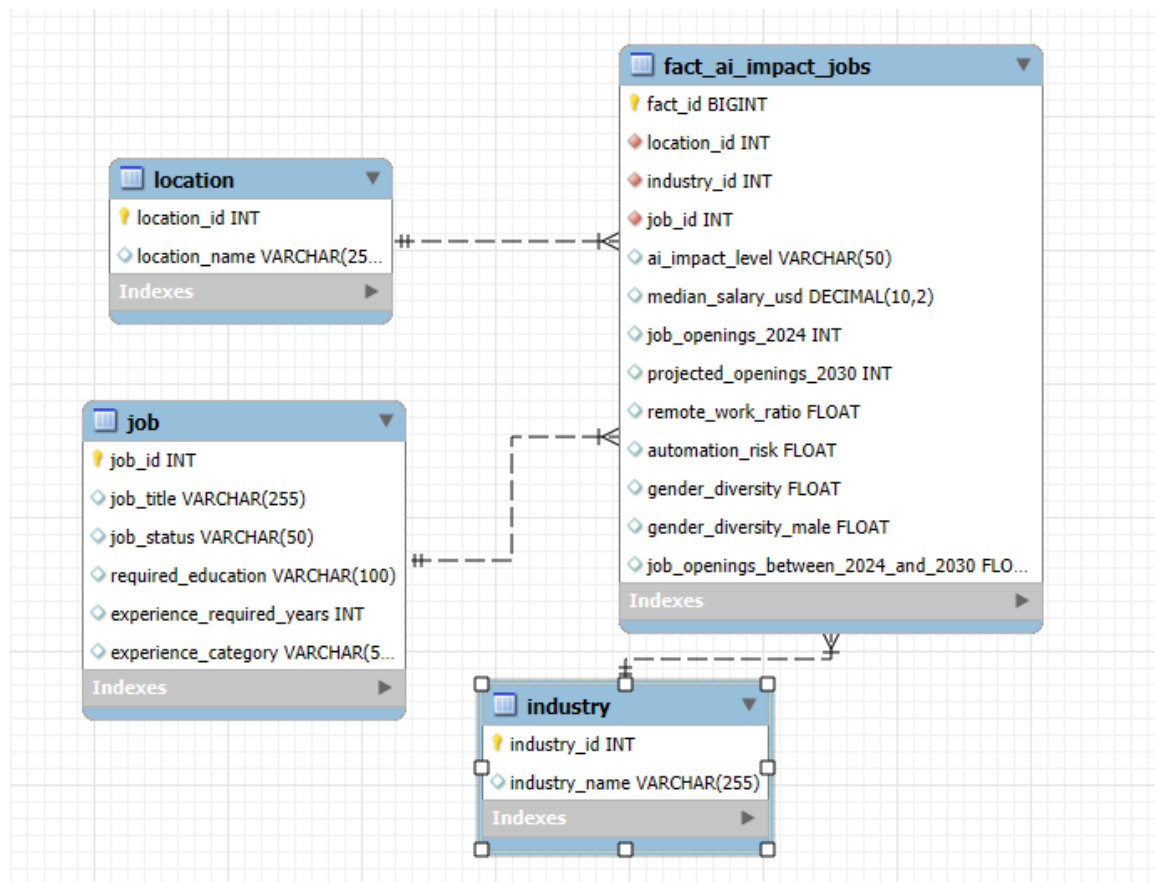


Excel Dashboard



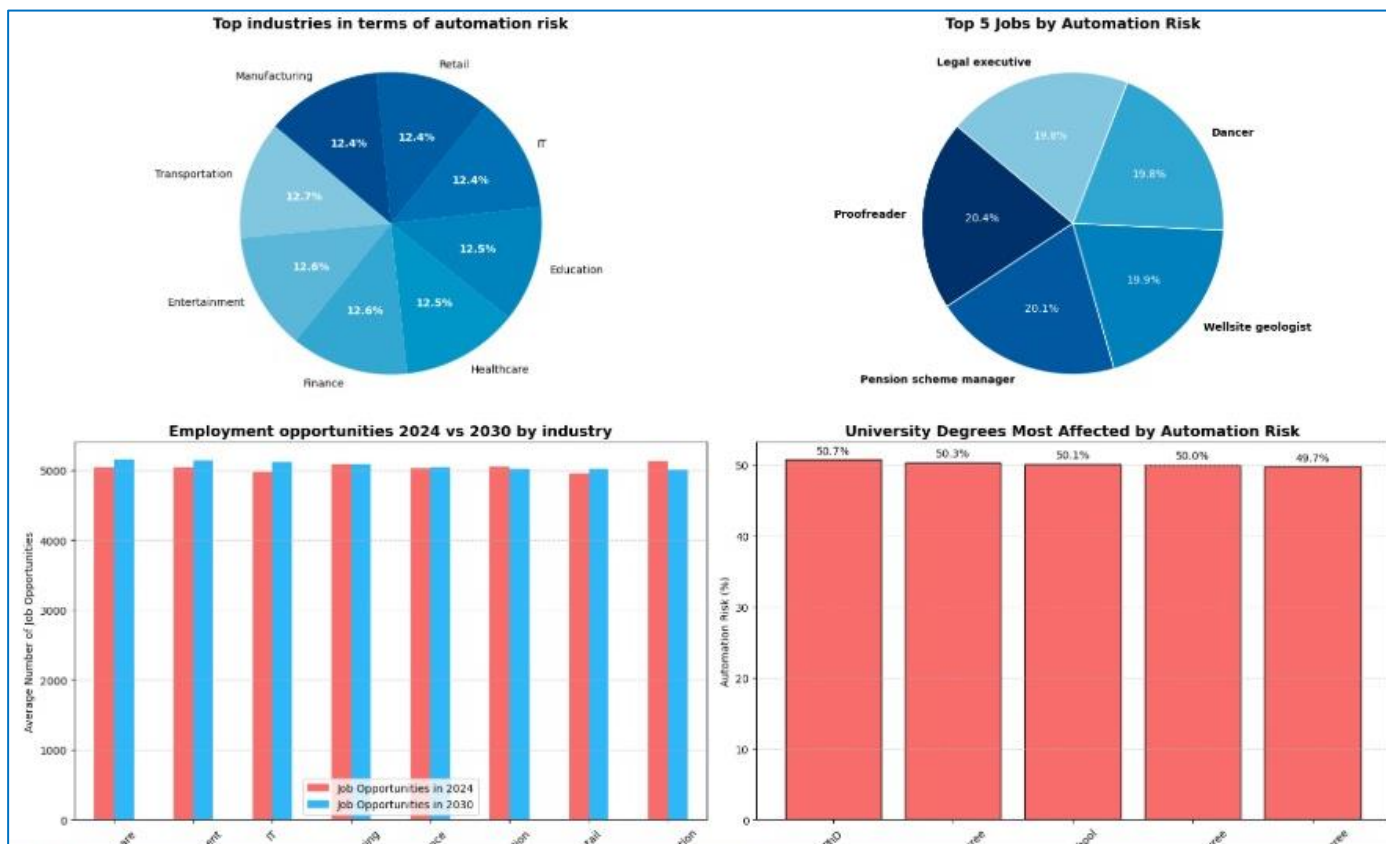


Data Modeling





Python Dashboard





DATASET ANALYSIS

What Happened? (Descriptive Analysis)

EDA Question: What industries show the highest exposure to the AI?

What will happen? (Predictive Analysis)

EDA Question: If current trends continue which jobs roles are most likely to grow or decline due to AI?

Why did it happen? (Diagnostic Analysis)

EDA Question: Which features (skills, education level, industry) most strongly predict job displacement or transformation?

What should we do? (Prescriptive Analysis)

EDA Question: What skills should workers learn to remain competitive?





EXPLORATORY DATASET ANALYSIS

What?

What is the dataset about?

Ex: Job roles, automation risks, industries, salaries.

Who?

Who does the data represent?

Ex: Employees, job categories, industries affected by AI

When?

When was this data collected?

Where?

Where does this data come from?

Kaggle.com



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DATA VISUALIZATION



TABLEAU





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DATA VISUALIZATION



THE AI IMPACT

WILL THE MACHINE REALLY
REPLACE HUMANS?

Market Forecasting

AI impact

Geographical

Project Members:
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Supervised by: DR. Amal Mahmoud



DATA VISUALIZATION

DEPI
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AI Impact On Job Market

Home

Market Forecasting

AI Impact

Geographical

Industry

(All)

Location

(All)

Education

(All)

Experience

(All)

Industry

8

Country

8

Job Title

639

Average Salary

90.12K

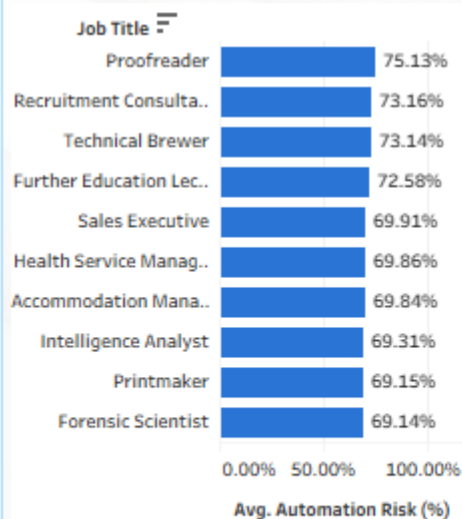
Job Openings (2024)

151.19M

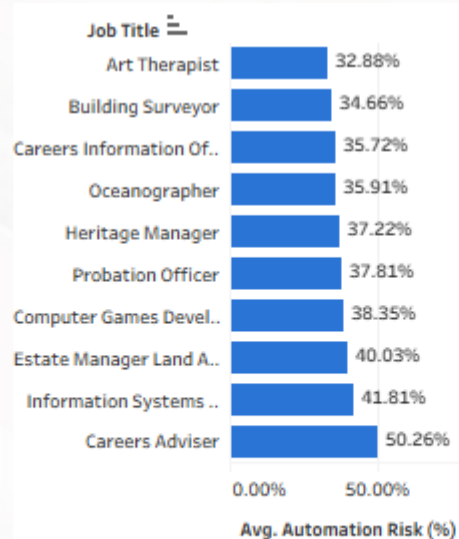
Projected Openings (2030)

152.23M

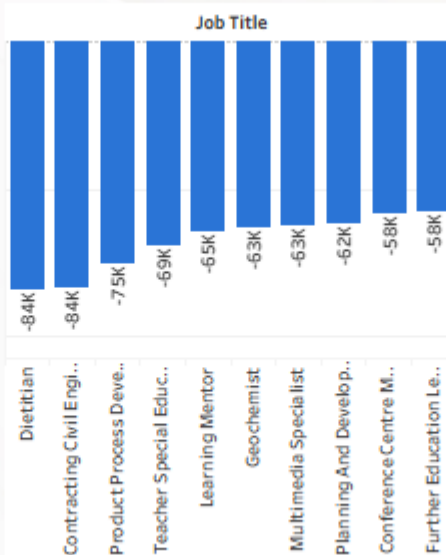
Top 10 Jobs AI Automation Risk



Least 10 Jobs AI Automation Risk



Jobs Decreasing By 2030



Forecast job market shifts by 2030

Industry	(2024)	(2030)
Entertainment	19.66M	20.04M
Manufacturing	19.59M	19.61M
Healthcare	19.01M	19.41M
Transportation	18.78M	18.34M
Education	18.75M	18.65M
Finance	18.73M	18.77M
Retail	18.35M	18.58M
IT	18.32M	18.83M



DATA VISUALIZATION





DATA VISUALIZATION



AI Impact On Job Market

Home

Market Forecasting

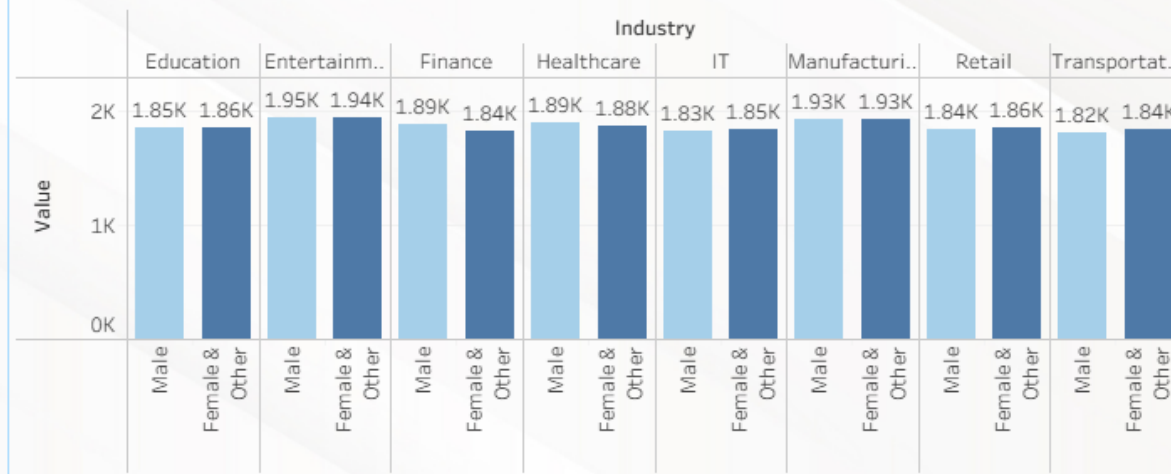
AI impact

Geographical

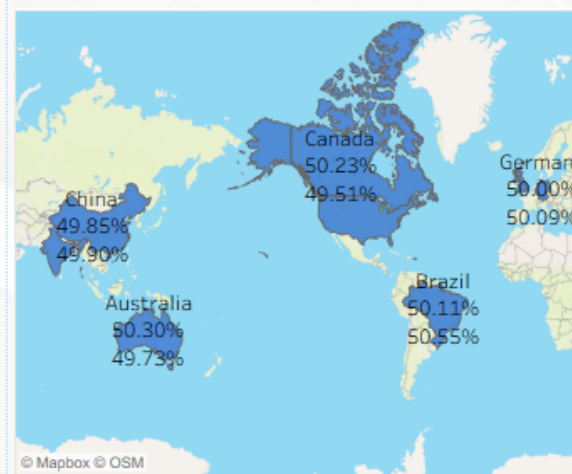
Industry (All) Location (All) Education (All) Experience (All)

Industry	Country	Job Title	Average Salary	Job Openings (2024)	Projected Openings (2030)
8	8	639	90.12K	151.19M	152.23M

Gender Diversity by Industry



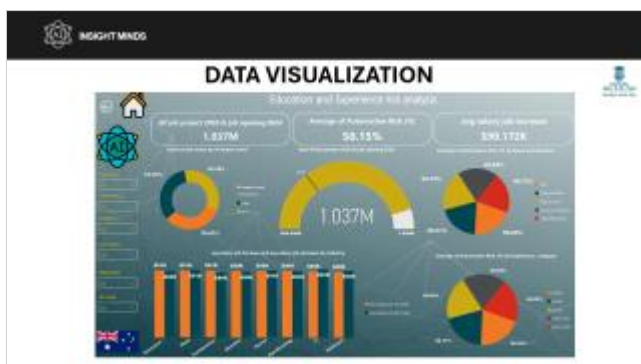
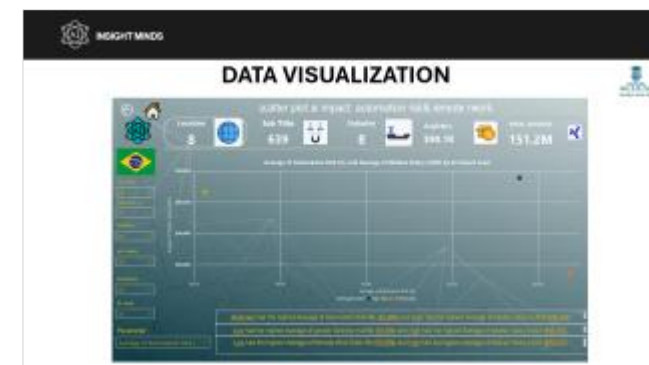
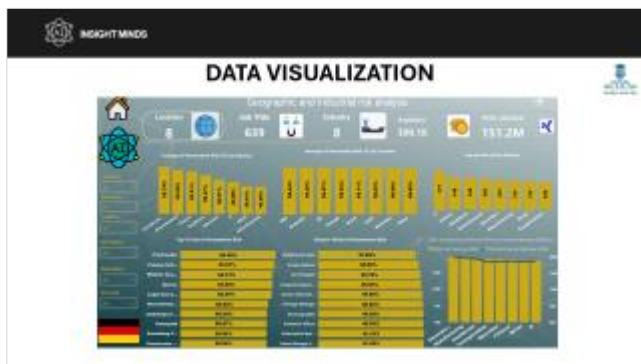
Automation Risk and Remote Work Across Location





DATA VISUALIZATION

Power BI





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DATA VISUALIZATION



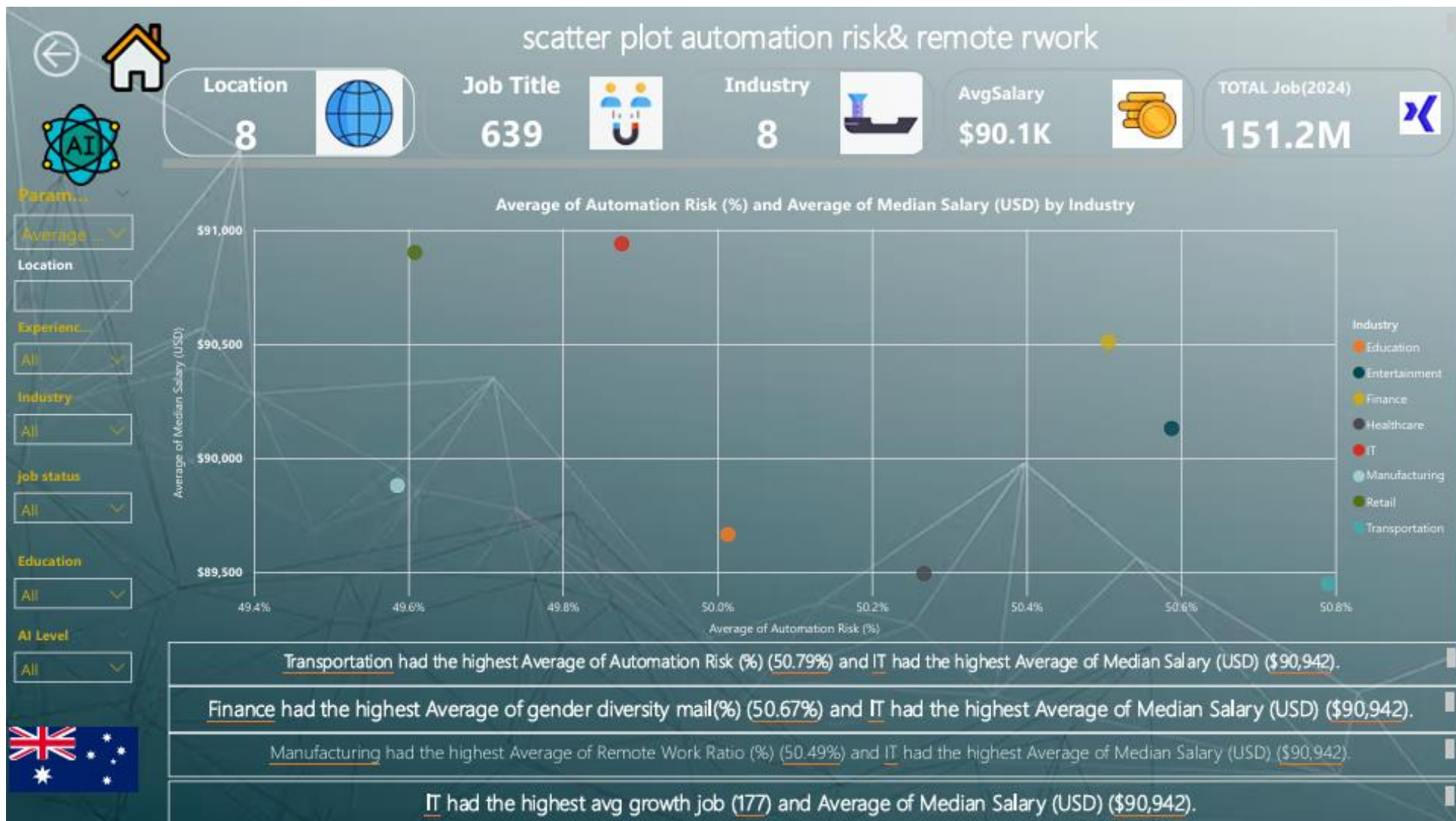


DATA VISUALIZATION



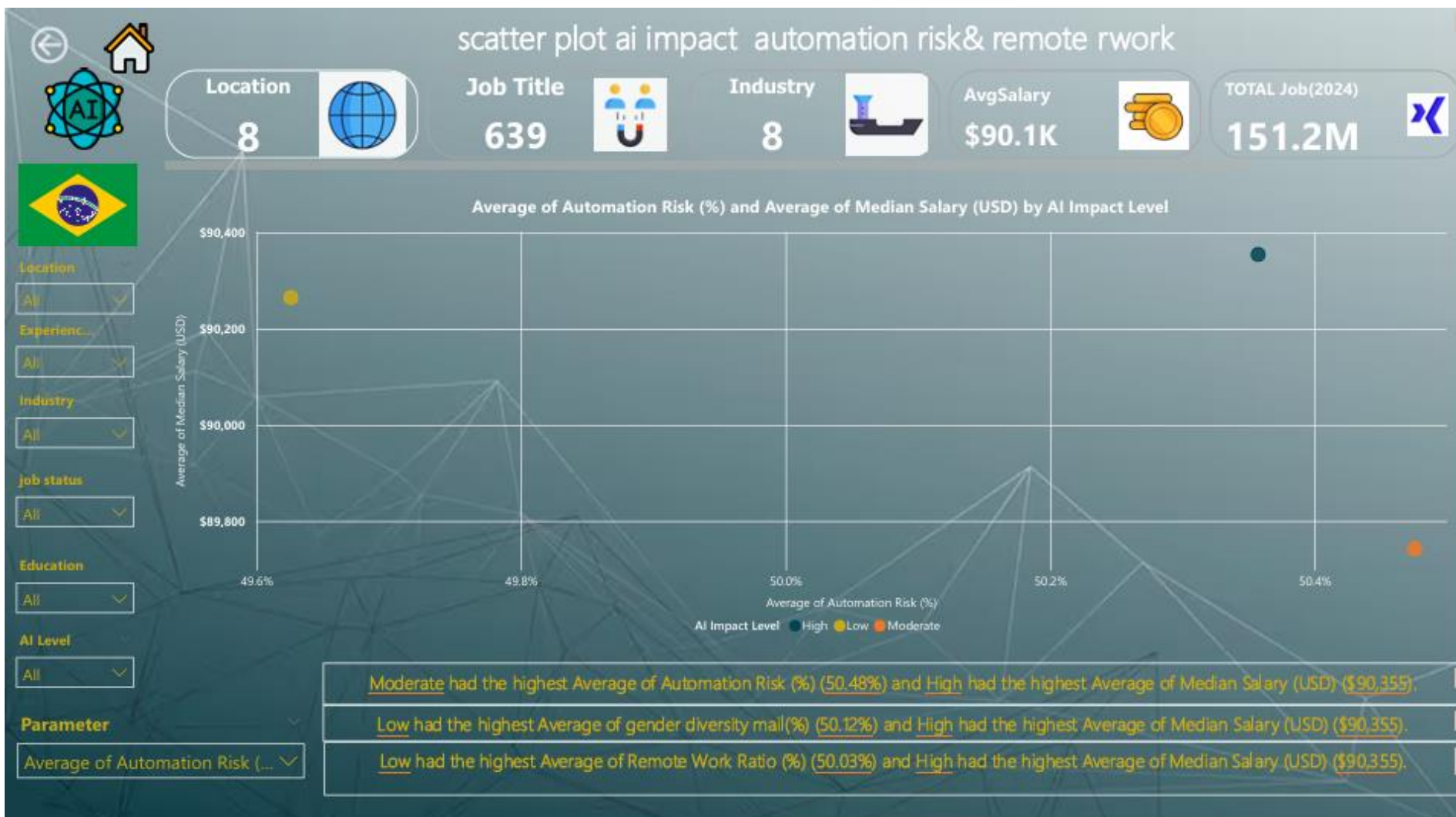


DATA VISUALIZATION





DATA VISUALIZATION





Conclusion

The analysis demonstrates that AI is reshaping the global job market through a balanced mix of automation, job shifts, and skill evolution. While certain job categories—particularly routine and operational roles—face high automation risk, the market remains stable overall, with projected openings showing only moderate change through 2030. Growth is concentrated in expert, supervisory, and high-skilled technical roles, reinforcing the increasing value of specialized expertise.

Geographically, automation exposure is consistent across countries, indicating that AI's impact is universal rather than region-specific. Industries such as entertainment, healthcare, education, and manufacturing are expected to experience steady growth, supported by evolving skill demands and technological integration. Overall, the findings highlight the need for continuous reskilling, stronger digital capabilities, and proactive market adaptation to ensure workforce readiness for the coming decade.



Main Outcomes

1. AI Automation Risk

The job market shows **balanced automation exposure**, with risk levels distributed almost evenly across Low (32.8%), Moderate (33.7%), and High (33.5%) categories.

Certain roles—such as **Proofreaders, Technical Brewers, Recruitment Consultants, and Sales Executives**—face automation risks above **70%**, indicating strong potential for AI-driven displacement.

In contrast, **creative, specialized, and interpersonal roles** (e.g., Art Therapists, Heritage Managers, Oceanographers) show significantly **lower automation risk**, mostly below **40%**.

2. Job Demand Forecast (2024–2030)

Total job openings are projected to grow from **151.19M in 2024** to **152.23M in 2030**, suggesting **stable but slow market expansion**.

Industries expected to grow most by 2030 include:

Entertainment (+0.38M)

Manufacturing (+0.02M)

Healthcare (+0.31M)

Education & Finance (moderate growth levels)



Main Outcomes

3. Jobs Decreasing by 2030

Several roles show noticeable decline in demand, including **Dietitians, Contracting Supervisors, Learning Mentors, and Manufacturing Specialists**, with job losses ranging from **58K to 84K**.

Declines are concentrated in roles with **routine, manual, or repetitive task structures**, reinforcing AI's impact on operational positions.

4. Experience-Level Outlook

Expert-level roles experience the largest growth (+583K), reflecting increased demand for innovation-driven and decision-making positions.

Junior-level roles decline significantly (-134K), highlighting automation's impact on entry-level and repetitive functions.

Supervisor and senior positions show moderate increases.



Main Outcomes

5. Geographic Insights

Automation risk is relatively consistent globally, ranging between **49% and 51%**, indicating AI's **universal market impact**.

Countries like **Canada, Brazil, Germany, China, and Australia** display similar adoption and automation exposure rates.

6. Gender Diversity

Gender distribution remains **balanced across industries**, with male and female participation showing minimal variance, indicating that AI-driven shifts are **industry-based rather than gender-based**.



Recommendations

1. Prioritize Reskilling and Upskilling

Invest heavily in training related to **AI literacy, data analytics, automation tools, and digital workflows**. Industries with high automation risk should shift toward **reskilling pathways** to preserve employment levels.

2. Support Transition for High-Risk Occupations

Develop targeted programs for roles exceeding **70% automation risk**, such as proofreading, sales, and administrative support. Offer structured transition plans into **emerging roles** with lower automation exposure.

3. Strengthen Talent Pipelines for High-Growth Segments

Expand training for roles seeing high demand growth. Encourage partnerships between governments, universities, and industries to align talent supply with market needs.



Recommendations

4. Enhance Workforce Planning Through Data

Organizations should use forecasting models to **predict skills shortages**, job shifts, and market gaps. Cluster analysis and predictive analytics can support smarter workforce planning.

5. Leverage AI as an Enabler, Not a Replacement

Encourage companies to adopt AI to **augment** employee productivity rather than replace roles. Emphasize hybrid workflows where AI supports repetitive tasks while humans focus on strategic, creative, and customer-facing work.

6. Prepare Policy and Regulatory Frameworks

Governments should introduce policies supporting safe, ethical, and transparent AI adoption. Incentives can help industries modernize without accelerating unemployment.



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Thank You
